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Molecular dynamics simulation of characteristics of the nanodroplet evaporation

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Abstract

The molecular dynamics method is used to simulate the evaporation process of nanodroplets on heated substrates by LAMMPS software. The simulation process includes three stages: the relaxation of argon droplets at 84K, heating the substrates to 200 K, keeping the substrates temperature at 200 K to observe the evaporation of the droplet. By measuring the change of the whole morphology of the droplets during the evaporation process, the changes of the edge thickness are obtained. By analyzing the atomic velocity and displacement, the trajectories of the particles at the edge of the evaporation process are revealed and distribution of the temperature of the droplets are expressed. It is concluded that when the substrates at 200K, the whole droplet evaporation process is conclude four stages: fixed evaporation stage, the stick slip motion stage, once again into the fixed evaporation stage and fixed contact angle and the contact radius are decreasing the evaporation process until the end. Finally, the experimental platform was set up to verify the validity of the simulation. Without considering the influence of plate roughness, the height variation of the droplet obtained from the experiment has a high consistency with the simulation results.

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Keywords: Solid wall heating; droplet evaporation; molecular dynamics; heat transfer; edge thickness.

1. Introduction

Droplet evaporation is a phenomenon that occurs spontaneously when the ambient pressure does not reach saturation, its evaporation at high temperature wall has important engineering value, such as the evaporation problem

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of the fuel in the combustion process of the engine cylinder, the cooling problem of the electronic device, material surface coating and so on. Since micro-scale and micro-time are challenged experimentally, the research on droplet evaporation on high-temperature wall mainly focuses on the macroscopic aspect, macroscopic physical quantities are based on the hypothesis of macro-continuity. When the diameter of the droplet reaches the nanometer, the continuity assumption will not be established. With the development of the computer technology, it is possible to study the micro-scale heat transfer phenomenon of evaporation of nano-droplets on the surface of thermosetting solid wall from the atom level. Molecular dynamics method is from the molecular level, by solving the Newton equation of motion of each molecule in the system and using the relevant simulation software to obtain the particle position, velocity, interaction force in the system and other data, through the relevant statistical theory the thermodynamic data of volume, temperature and pressure under the corresponding macroscopic conditions were obtained by Lv Jiazhen et al [1] in this method in the surface and interface, complex fluid, supercritical fluid and membrane applications in the field were reviewed. Many scholars use molecular dynamics method to study the wetting dynamics of droplets under normal temperature environment and the factors that affect their evaporation [2-4]. The effect of the substrate characteristics on the evaporation of the droplets was discussed without considering the wettability of the droplets on the solid hot substrate and the evaporation characteristics [5, 6]. Pan Yi et al[7] simulated the properties of thin liquid film was composed of argon at 150K, 300K and the metal substrate dropped from high temperature to 110K, to analyze the evaporation of liquid films and the reconstruction of liquid films process. Hou Zhaolong et al [8] simulated the behavior of suspended nano-droplets and the factors that affect their evaporation, which provided a good directional guidance for the future simulation.

In this paper, the non-equilibrium molecular dynamics method was used to simulate the evaporation process of nano-argon droplets on the metal platinum substrate by Large-scale Atomic/Molecular Massively Parallel Simulator （LAMMPS）. The changes of the thickness and temperature of the droplet were obtained during the process. From the molecular level to reveal the droplet internal heat transfer law. Through the comparison with the droplet evaporation process in macroscopic state, the change of droplet edge morphology was observed and the corresponding variation rule was obtained.

2. Model and method

2.1. Molecular dynamics simulation

Molecular Dynamics Simulation (MDS) is based on the basic research objects of molecules (atoms or other ions, conveniently referred to as molecules), and considers the system as a collection of large numbers of molecules, using classical mechanics or quantum mechanics study the laws of movement of molecules, and finally get the macroscopic nature of the system and the basic laws. According to the laws of classical mechanics, the force of each atom in the system is the potential energy gradient of the atom, the acceleration is calculated by classical Newtonian mechanics, and then the time is integrated to obtain the new coordinates and velocity of each molecule after the specified integral step. After a period of integration steps, each molecule has its own trajectory, by using Visual Molecular Dynamics、Open Visualization Tool [9] and other visualization software based on the trajectory of the system to obtain the overall structure, thermodynamics, dynamics, energy and other aspects of the analysis, and ultimately get the calculation result of the relevant physical quantity.

2.2. Physical model

In this simulation, argon is used as the droplet medium. In order to simulate the droplet evaporation state and avoid the influence of the boundary effect, the reliability of the calculation is ensured, and the periodic boundary condition is adopted in the X and Y directions, in the Z direction using free boundary conditions to calculate the changes number of atoms in the evaporation process. The simulated initial physical model cube structure and the area size is 30 nm x 30 nm x 50 nm, including two regions: droplets consisting of argon atoms and substrate with metal platinum atoms arranged in the FCC is placed at the bottom of the simulation region. The force between the platinum atoms is the harmonic force (spring force), and the distance between the adjacent atoms is 2.77 nm, The elastic constant is 46.8 N/m. The bottom two layers of the bottom substrate are fixed to prevent deformation of the system during the

simulation. Lyle N. Long et al [10] pointed out that when there is no pre-existing configuration the atoms are arranged as a solid-like face-centered-cubic (FCC) lattice. So, the composition of the 6,000 argon atoms in the liquid area with FCC. The arrangement is placed in the center of the substrate. The top of the simulated box is far enough away from the substrate to avoid affecting the droplet evaporation process.

The velocity components of atoms along the three coordinate directions are assigned randomly before the simulation starts and then the velocities are rescaled according to the system temperature. Therefore, the initial velocity of the atoms is randomly selected by the set initial temperature of 84 K according to the Maxwell-Boltzmann distribution. The Lennard Jones 12-6 potential model was used to describe the effect between argon atoms, as Eq. (1) expresses：

$$U(r_{ij}) = \begin{cases} 4\varepsilon\left[\left(\frac{\sigma}{r_{ij}}\right)^{12} - \left(\frac{\sigma}{r_{ij}}\right)^{6}\right], & r_{ij} \le r_c \\ 0\,, & r_{ij} > r_c \end{cases} \tag{1}$$

where ε is potential energy constant, σ is the collision diameter and r is the intermolecular distance.

The embedded atom model(EAM) was considered for the interaction between the surface Pt atoms. EAM based models are particularly useful for metallic systems and based on this model, the total energy is given by Eq. (2) expresses.

$$E_t = \sum F_i(\rho_i) + \frac{1}{2}\sum_{i\neq j} U_{ij}(r_{ij}) \tag{2}$$

where ρ_i is the host electron density at atom i due to the remaining atoms of the system, $\sum(\rho_i)$ is the energy required to embed atom i into the background electron density ρ and $U(r_{ij})$ is the pair potential between atoms i and atoms j separated at distance r_{ij}.

The interaction between Pt and Ar atoms can be expressed according to Lorentz-Brethelot[11] rule of combination, the dimension and potential interation parameters $\sigma_{ar\text{-}pt}$ and $\varepsilon_{ar\text{-}pt}$ can be calculated by Eq. (3), respectively.

$$\begin{cases} \sigma_{ar-pt} = \frac{1}{2}(\sigma_{ar} + \sigma_{pt}) \\ \varepsilon_{ar-pt} = \sqrt{\varepsilon_{ar}\varepsilon_{pt}} \end{cases} \tag{3}$$

The physical parameters of Pt and Ar atoms are listed in Table 1.

Table 1. Physical parameters of atoms.

Parameters	m/Kg	Potential energy constant ε/J	Collision diameter σ/m
Ar	6.636×10^{-26}	1.67×10^{-21}	3.405×10^{-10}
Pt	32.39×10^{-26}	0.835×10^{-21}	2.774×10^{-10}
Ar-Pt	/	1.181×10^{-21}	3.089×10^{-10}

3. Results and discussion

The simulation process is divided into three stages. The first stage is in the entire simulation area to obtain three phases, namely, the liquid droplet surrounded by its gas and the solid substrate. In the regular system, the speed calibration method is used to balance the temperature of the droplet and the substrate at 84K. The time step is 0.01ps, which is called the relaxation process. It can be concluded from the temperature change curve in the process that the temperature of the substrate and liquid phase in the simulation domain reaches the desired temperature after a small degree of oscillation. At this temperature, the liquid argon atoms are naturally relaxed after 500ps to form hemispherical droplets in the center of the substrate.

The second stage is the temperature-raising evaporation process, the time step is 0.005ps, at first keep the liquid region in the micro-regular system without the energy exchange with the outside, note that thermostat was only applied to the substrate atoms to 200K with the time of 150ps. After the end of the heating process, the whole system uses the regular system to heat the droplets to observe its evaporation. From the temperature diagram can be seen in the solid wall heating stage because the droplets in the micro-regular system under the current and no energy exchange, but because of the formation of droplets of argon atoms own vibration, so the droplet temperature will be slightly shock, this stage the duration of the process is 150ps, and the droplet is heated after the end of the wall temperature rise. After 150ps, the droplet temperature rises from the initial 84K to 110K.

The third stage is that the NVT ensemble was used, where the Berendsen thermostat was chosen to control substrate temperature at 200K and heat the droplets to observe its evaporation. Fig.1 shows the trend of droplet height H and contact line length D over time during evaporation. Fig. 2 shows the representation of the droplet morphological parameters. Fig.3 shows the evolution of the profile of the droplets over time during the entire evaporation process with the substrate temperature maintained at about 200 K.

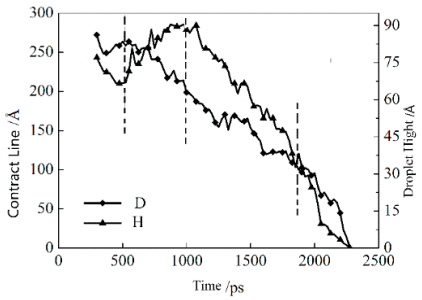


Fig. 1. Relationship between droplet D and H with time.

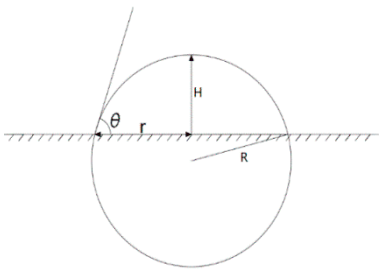


Fig. 2. Schematic diagram of droplet morphological parameter characterization.

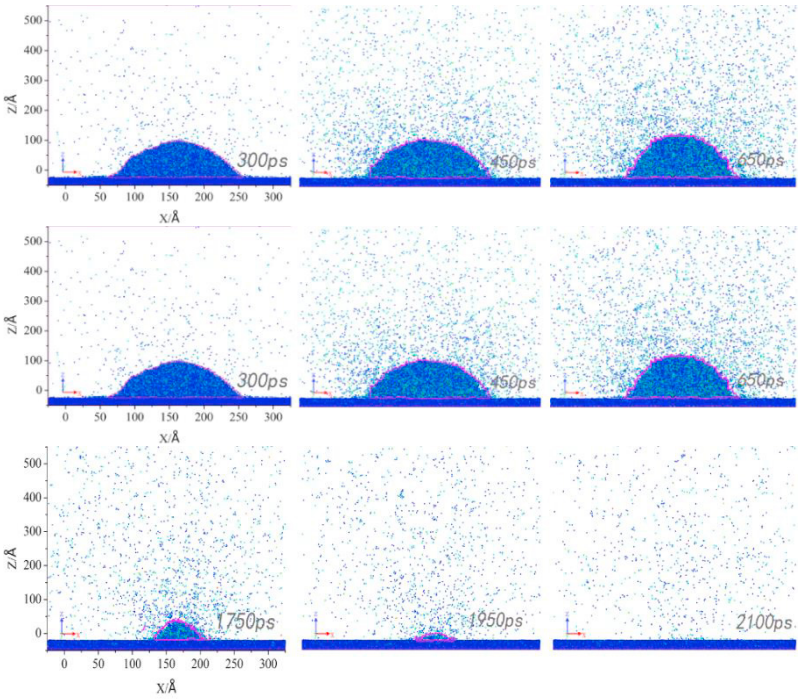


Fig. 3. Molecular distribution of droplet evaporation process.

Combining Fig. 1 and Fig. 3 shows that the entire evaporation process lasts 2200 ps. The droplet has a natural spreading process on the metal substrate due to the wetting effect in the substrate heating stage [3], so that the droplet height and the relaxation phase are slightly reduced at the beginning of the evaporation process. It can be seen from Figure 3 argon droplets in the 200K hot metal substrate on the evaporation process is more complex, the overall can be divided into four stages. The first stage: droplet fixation in the base evaporation, that is, the contact line length is basically the same, the droplet height continues to decrease. The second stage of sticky slip contact line length suddenly decreases, the droplet height suddenly becomes larger, the corresponding contact angle θ fluid suddenly increased. The third stage re-enters the fixation phase: that is, the contact line is fixed, the droplet height continues to decrease, and the corresponding contact angle is also continued to decrease. In this stage, the contact line has two drop points. The fourth stage droplet contact line decreases with the droplet height, and the corresponding contact angle is reduced until the evaporation stage ends.

4. Experiment

Fig. 4 shows the evaporation process of 30uL droplets on a 353 K hot metal aluminium plate. After several tests to verify, 30uL droplets at rest diameter 11.56~12.05mm, in the course of the test when the diameter of the droplets at rest 11.84mm.

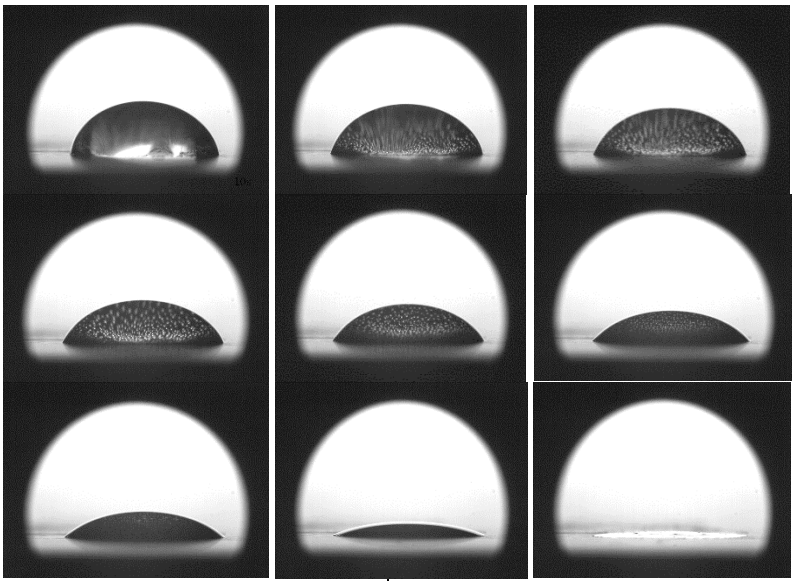


Fig. 4. Droplet evaporation experiment process.

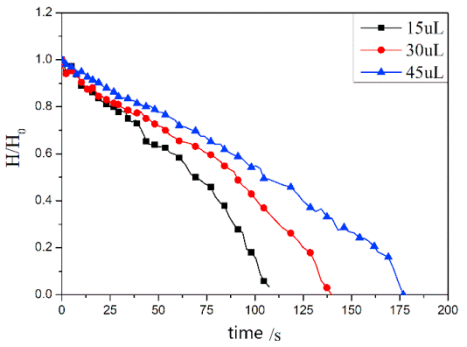


Fig. 5. The change in height of different volume droplets at 80°C.

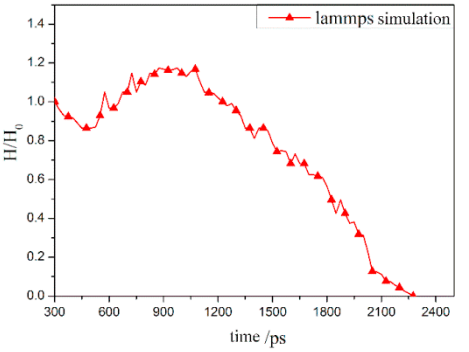


Fig.6 Drop height variation under lammps simulation conditions.

The H is the change in droplet height with time, H_0 is the height of the initial droplet, and the dimensionless number H/H_0 is used to represent the droplet change of the droplets height with the evaporation time. Fig.5 shows that as the volume of the droplet increases, the evaporation time gradually increases. It can be seen by comparing with Fig.6 that the height of the droplet of the experiment is changed without considering the influence of the roughness of the plate, there is a high consistency between experimental results and the simulation results. (0-40s), the droplet evaporates slowly in the early stage, and the droplet height and droplet contact line change relatively small; (40-60s) there is a short stick-slip movement: the droplet height becomes larger suddenly and the contact line becomes smaller; (60-100s) evaporation relative to the first phase is faster, the droplet height quickly becomes smaller, the line of contact changes less; (100-150s) Evaporation into the latter part of the rate of speed up, the contact line quickly becomes smaller until 0, the drop height is rapidly reduced to 0.

5. Conclusion

In this paper, the molecular dynamics simulation method was used to simulate the evaporation process of nano-droplets on metal heating substrates by LAMMPS software. The whole simulation process included three stages: the whole system was relaxed at 84K, heating the substrate to 200K, keep the substrate temperature unchanged so that the droplets warm up to observe its evaporation. The changes of droplet temperature and edge thickness during this process were observed and analysed. The following conclusions are drawn:

(1) The evaporation process of argon droplets on the hot metal substrate is complicated, and the whole process is divided into four stages: the droplet solidification stage, the viscous slip stage, the droplet solidification evaporation stage and the contact angle decrease with the contact radius until the end of evaporation.
(2) By analysing the changes of the droplet height of the experiment without considering the influence of the roughness of the plate. There is a high consistency between experimental results and the simulation results.

Acknowledgements

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